

where $\pi(i)$ and $\sigma(i)$ are the indices of the i -th position in the ranking method and the ideal attention-based ranking, respectively. The ALR for each head and layer is then calculated as $Y_{l,h} = \sum_{m=1}^n (L_{l,h}^m - L_{l,h,\text{ref}}^m)$.

A lower $Y_{l,h}$ indicates that the ranking method closely approximates the ideal attention-based compression.

5. **Aggregation** We repeat the above steps for multiple prompts and average the ALR values to obtain the final ALR matrix across layers and heads.

F Metrics and Prompts

F.1 String Match Score

The string matching score is calculated as:

$$\text{String Matching Score} = \frac{\text{Number of correctly matched characters in predicted string}}{\text{Total number of characters in GT}} \times 100$$

F.2 GPT-4-Judge Prompt

For the GPT-4-Judge metric used in evaluating free response question answering tasks, we accessed the GPT-4o model through OpenAI's API.

For the GPT4-Rouge metric, the prompt given to the model is:

You are shown ground-truth answer(s) and asked to judge the quality of an LLM-generated answer. Assign it a score from 1-5 where 1 is the worst and 5 is the best based on how similar it is to the ground truth(s).
Do NOT explain your choice. Simply return a number from 1-5.

```
====GROUND TRUTHS====
{labels}
```

```
====ANSWER====
{prediction}
```

For the other three GPT4-Judge based on criteria, the prompt given to the model is:

You are shown a prompt and asked to assess the quality of an LLM-generated answer on the following dimensions:

```
===CRITERIA===
{criteria}
```

Respond with "criteria: score" for each criterion with a newline for each criterion. Assign a score from 1-5 where 1 is the worst and 5 is the best based on how well the answer meets the criteria.

```
====PROMPT====
{prompt}
```

```
====ANSWER====
{prediction}
```

The list of criteria is:

```
CRITERIA = {
  "helpful": "The answer executes the action requested by the prompt without extraneous detail.",
  "coherent": "The answer is logically structured and coherent (ignore the prompt).",
  "faithful": "The answer is faithful to the prompt and does not contain false information.",
}
```